MODULE II • LECTURE 02

Deep Learning for NLP

Activation Functions

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Sigmoid • Tanh • ReLU • Leaky ReLU • Softmax

FROM WEIGHTED SUMS → NONLINEAR LEARNING

Recap: The Artificial Neuron

From the artificial neuron to the activation step.

NEURON

$z = w^T x + b$

$a = f(z)$

Last lecture: inputs, weights, bias, layers and forward propagation.

TODAY

The focus is $f(z)$:

How should a neuron transform its weighted sum?

Learning Objectives

What students should be able to explain and calculate.

BY THE END OF THE LECTURE

- Explain why nonlinearity is needed.
- Calculate sigmoid, tanh and ReLU outputs.
- Explain softmax.
- Choose suitable activations for layers.
- Explain vanishing gradients and dying ReLU.
- Connect activation derivatives to backpropagation.

Why Do We Need an Activation Function?

Why weighted sums alone are not enough.

WITHOUT ACTIVATION

$a = w^T x + b$

Only a linear/affine transformation.

WITH ACTIVATION

$a = f(w^T x + b)$

Nonlinearity changes what the network can represent.

Nonlinearity makes stacked layers more expressive.

The Problem with Only Linear Layers

A network made only of linear transformations remains linear.

Linear → Linear → Linear

can be collapsed into another linear/affine transformation

MATHEMATICAL INTUITION

$$W_3(W_2(W_1x+b_1)+b_2)+b_3 = W'x+b'$$

Add Nonlinearity

Nonlinearity enables multilayer networks to model complex functions.

INPUT

X

HIDDEN

Linear + Nonlinear

OUTPUT

Complex nonlinear function

The activation breaks the simple linear composition.

What Is an Activation Function?

Activation transforms the weighted sum.

$$z = w^T x + b$$

$$\downarrow$$

$$a = f(z)$$

ROLE

The activation transforms the weighted sum into an output signal.

Activation Function Families

Common activation functions and their roles.

STEP

Classical perceptron

Hard 0/1 decision

Not suitable for ordinary gradient training.

SIGMOID / TANH

Sigmoid: 0–1
Tanh: -1–1

Smooth but can saturate.

RELU / SOFTMAX

ReLU: hidden layers
Softmax: multiclass output

Sigmoid Function

Sigmoid maps a real-valued score to 0–1.

$$\sigma(z) = 1 / (1 + e^{-z})$$

RANGE

$0 < \sigma(z) < 1$

USE

Binary probability

INTERPRET

Larger z → output closer to 1

Sigmoid Behavior

How sigmoid behaves for different inputs.

$z \ll 0$

$\sigma(z) \to 0$

Strongly negative score.

$z = 0$

$\sigma(0) = 0.5$

$z \gg 0$

$\sigma(z) \to 1$

Strongly positive score.

Sigmoid smoothly converts an unrestricted score into a bounded value.

Sigmoid Example

A numerical sigmoid calculation.

GIVEN

$z = 2$

CALCULATION

$\sigma(2) = 1 / (1 + e^{-2})$
≈ 0.881

The neuron outputs approximately 0.881.

Sigmoid for Sentiment Classification

Using sigmoid for binary sentiment classification.

NETWORK OUTPUT

$z = 2.4$

$\sigma(2.4) \approx 0.917$

DECISION

$P(\text{Positive}) \approx 0.917$

$\text{Threshold} = 0.5$

→ Positive

Sigmoid Derivative

Why the sigmoid derivative matters for learning.

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

MIDDLE OF CURVE

Derivative is relatively large.

SATURATED REGION

Derivative becomes very small → vanishing-gradient risk.

Tanh Function

Tanh produces zero-centered outputs.

$$\tanh(z) = (e^z - e^{-z}) / (e^z + e^{-z})$$

RANGE

$-1 < \tanh(z) < 1$

CENTER

$\tanh(0) = 0$

PROPERTY

Zero-centered

Sigmoid vs Tanh

Tanh versus sigmoid.

SIGMOID

Range 0–1
Zero-centered: No
Probability interpretation: Yes
Saturation: Yes

TANH

Range -1–1
Zero-centered: Yes
Probability interpretation: No
Saturation: Yes

Tanh Example

A numerical tanh interpretation.

$z = 0$

$\tanh(0) = 0$

LARGE VALUES

$z \to +\infty \Rightarrow \tanh(z) \to 1$
$z \to -\infty \Rightarrow \tanh(z) \to -1$

ReLU

ReLU: Rectified Linear Unit.

$$\mathrm{ReLU}(z) = \max(0, z)$$

IF $z < 0$

$\mathrm{ReLU}(z) = 0$

IF $z \geq 0$

$\mathrm{ReLU}(z) = z$

ReLU Examples

ReLU examples.

-3

ReLU = 0

-1

ReLU = 0

2

ReLU = 2

5

ReLU = 5

Negative input → 0 • Positive input → passes through

Why Is ReLU Popular?

Why ReLU became a common hidden-layer activation.

SIMPLE

max(0,z)

Very cheap to compute.

SPARSE

Negative inputs become zero.

PRACTICAL

Often works well for deep hidden layers.

ReLU is a common default activation for hidden layers.

The Dying ReLU Problem

The dying-ReLU problem.

NEGATIVE REGION

For $z < 0$:
$\mathrm{ReLU}(z) = 0$
$\mathrm{ReLU}'(z) = 0$

CONSEQUENCE

A neuron may become effectively inactive if it receives mostly negative inputs.

This is called the "dying ReLU" problem.

Leaky ReLU

Leaky ReLU keeps a small negative slope.

$$\text{Leaky ReLU}(z) = \begin{cases} \alpha z & \text{if } z < 0 \\ z & \text{if } z \geq 0 \end{cases}$$

NEGATIVE SIDE

A small slope remains.
Example $\alpha = 0.01$

BENEFIT

Negative inputs are not completely discarded.

Why Do We Need Softmax?

Softmax converts class scores into probabilities.

TASK

Sports
Business
Politics
Technology

NEED

Convert several raw class scores into probabilities that sum to 1.

Softmax Formula

Softmax mathematical formulation.

$$P(y=i) = e^{z_i} \;/\; \Sigma_j\, e^{z_j}$$

K

Number of classes

z_i

Score for class i

PROPERTY

Probabilities sum to 1

Softmax Example

A multiclass example.

RAW SCORES

Sports 2.0
Business 1.0
Politics 0.2
Technology 0.5

APPROX. PROBABILITIES

Sports ≈ 0.61
Business ≈ 0.22
Politics ≈ 0.10
Technology ≈ 0.07

Largest probability → predicted class: Sports

Choosing the Activation Function

Choosing an activation by layer and task.

HIDDEN

Usually ReLU

Nonlinear feature transformation.

BINARY OUTPUT

Usually Sigmoid

Positive-class probability.

MULTICLASS OUTPUT

Usually Softmax

Probability distribution over classes.

Activation Functions in an NLP Network

Activation functions inside an NLP classifier.

INPUT

TF-IDF / Embedding

HIDDEN 1

Linear → ReLU

HIDDEN 2

Linear → ReLU

OUTPUT

Linear → Sigmoid

Nonlinear hidden layers allow the classifier to learn feature combinations.

Activation Functions and Backpropagation

Activation derivatives participate in backpropagation.

$$\partial L/\partial w = (\partial L/\partial a)(\partial a/\partial z)(\partial z/\partial w)$$

CHAIN RULE

The activation derivative is part of the gradient.

WHY IT MATTERS

Poor gradient behavior can make learning slow.

Activation Function Comparison

Quick comparison of major activation functions.

STEP	0 / 1	Perceptron
SIGMOID	0–1	Binary output
TANH	-1–1	Zero-centered
RELU	0–∞	Hidden layers
SM	Σ = 1	Multiclass

Classroom Check

Short classroom check.

Q1

Which is common in hidden layers?

A. Sigmoid
B. ReLU
C. Softmax

Q2

For binary sentiment output?

A. Sigmoid
B. Softmax
C. ReLU

Q3

Why is nonlinearity necessary?

Stacked linear layers remain linear.

Summary & Next Lecture

Summary and next lecture.

CORE IDEA

Activation functions introduce nonlinearity.

$a = f(w^T x + b)$

KEY CHOICES

Hidden → ReLU
Binary output → Sigmoid
Multiclass output → Softmax
Tanh → zero-centered alternative

NEXT LECTURE

Dropout & Network Design Principles — capacity, overfitting and regularization.